# Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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**Abstract**

A $d$-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ has *separable rank* at most $N$ if it is a sum of $N$ coordinatewise products $\sum_{n=0}^{N-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$. On a $d$-dimensional Cartesian grid of support width $L$, a rank-$R$ kernel evaluates in $dRL$ one-dimensional taps per grid point rather than $L^d$. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or band-limited wave propagators—with finite coordinatewise separable rank relies on signed or oscillatory profiles built from complex-rate Gaussians and exponential polynomials.

We give a complete machine-checked characterization in Lean 4 (with Mathlib) of the $d$-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) separableRank$(K) < \infty$ over $\mathbb{R}$ if and only if $\varphi$ agrees on $[0, \infty)$ with a well-formed exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$ (nonzero polynomials $P_i$, distinct rates $\lambda_i$ with $\mathrm{Im}\,\lambda_i \geq 0$, and $P_i \in \mathbb{R}[X]$ when $\mathrm{Im}\,\lambda_i = 0$); (2) over $\mathbb{C}$, a continuous radial profile $\psi(\|x\|^2)$ has exact $d$-way separable rank $\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$ for nonzero polynomials $P_\mu$ at distinct rates; (3) over $\mathbb{R}$, a continuous radial profile $\varphi(\|x\|^2)$ has exact $d$-way separable rank $\sum_i c_d(P_i, \lambda_i)$ where each real-rate component ($\mathrm{Im}\,\lambda_i = 0$) costs $(d-1)\deg P_i + 1$ real $d$-way products (achieved by real Lagrange interpolation) and each nonreal conjugate-pair component ($\mathrm{Im}\,\lambda_i > 0$) costs $R_i = 2(d-1)\deg P_i + d$ real $d$-way products, achieved via an alternating $R_i$-point root-of-unity polynomial-product identity (giving $d$ rather than $2^{d-1}$ real products for a complex Gaussian when $\deg P_i = 0$) and proved optimal by Alder–Strassen degree peeling over $\mathbb{R}$; and (4) in $d = 2$, scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ preserves rank unconditionally, whereas for $d \geq 3$ it can strictly lower rank (e.g. $\mathrm{Re}(e^{i\|x\|^2})$ has complex rank 2 and real rank $d$, and $\|x\|^2\cos(\|x\|^2)$ in $d = 3$ has complex rank 6, real border rank at most 6, and exact real rank 7); moreover, in $d = 2$ the exact rank equals the degree of the minimal monic constant-coefficient half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$ ($p \in \mathbb{C}[X]$).

Necessity reduces to showing that a continuous one-parameter semigroup $M : [0, \infty) \to \mathcal{A}$ is an exponential $M(t) = \exp(tA)$. Alongside the 96-line Banach-algebra integral-smoothing proof used in the main development, we formalize four finite-dimensional alternatives—one by finite differences in Lean 4 and three by Euler's infinitesimal step in Lean 4 and Isabelle/HOL using a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ that avoids any *a priori* difference-quotient bound—and compare their lengths, library reliance, and generality. Finally, we connect the root-of-unity identity to two-dimensional harmonic ("beach-ball") differential operators and evaluate concrete pass-count reductions for steerable derivative filter banks and three-dimensional isotropic propagators.

# 1 Introduction

Convolving a $d$-dimensional array with a dense $L \times \cdots \times L$ spatial kernel requires $L^d$ multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-$R$ coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the $d$-dimensional convolution reduces to $R$ cascades of $d$ one-dimensional convolutions, costing $dRL$ operations per sample (or $O(dR)$ operations independent of $L$ when the one-dimensional factors admit recursive causal/anti-causal realizations). The least $R \in \mathbb{N} \cup \{\infty\}$ for which (1) holds is the *separable rank* of $K$: matrix rank when $d = 2$, and functional canonical polyadic (CP) tensor rank when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

Maxwell's characterization of the normal distribution implies that a continuous nonvanishing radial kernel of separable rank 1 has a Gaussian profile $\varphi(s) = \alpha e^{\lambda s}$. Real-rate Gaussians ($\lambda < 0$), however, are strictly monotone and convex in $s = \|x\|^2$; nonnegative combinations of them cannot exhibit flat passbands, sharp cutoffs, or zero crossings. In image processing, optics, and numerical wave propagation, O. Niemitalo [13, 14] and related filter-design work [8] approximate non-Gaussian radial profiles—such as circular defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\mathrm{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe rings while keeping finite separable rank.

This raises three mathematical questions:

1. **Classification ($d \geq 2$):** Must *every* continuous radial profile $\varphi : [0, \infty) \to \mathbb{R}$ of finite $d$-way separable rank be an exponential polynomial in $s = \|x\|^2$?
2. **Exact $d$-way rank over $\mathbb{C}$ and over $\mathbb{R}$:** Given an exponential-polynomial or multi-Gaussian profile with distinct rates, what is the exact minimum number of $d$-way product terms in (1)? Can algebraic recombinations beat naive real-and-imaginary expansion, and does complexification preserve separable rank beyond two dimensions?
3. **Proof architecture in interactive theorem provers:** In formalizing necessity—whose analytic core is the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0, \infty)$—can an elementary infinitesimal proof in the style of Euler be made simpler than classical integral smoothing once formalized in Lean 4 or Isabelle/HOL?

## 1.1 Contributions and overview

All numbered theorems, propositions, and lemmas in Sections 2–4 are machine-checked without unproved steps (`sorry` in Lean 4, oracles in Isabelle/HOL) in the open-source repository `kernel-rank-lean`, using Lean `v4.36.0-rc2` with Mathlib `v4.36.0-rc2` and Isabelle2026; Section 5 (along with the border-rank remark in Section 3) develops analytical corollaries for harmonic differential operators and continuous-domain pass schedules:

- **Unified $d$-dimensional classification (Theorem 3.3, `separableRank_radial_lt_top_iff_exists`):** For every $d \geq 2$ and every $\varphi : \mathbb{R} \to \mathbb{R}$ continuous on $[0, \infty)$, the radial kernel $x \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)$ has finite $d$-way separable rank if and only if $\varphi$ agrees on $[0, \infty)$ with a canonical real exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$.

- **Exact $d$-D complex radial rank (Theorem 3.1, `separableRank_radial_complex_eq_iff`):** For $d \geq 2$ and $\psi : \mathbb{R} \to \mathbb{C}$ continuous on $[0, \infty)$, separableRank$(x \mapsto \psi(\|x\|^2)) = N$ if and only if $\psi(s) = \sum_{\mu \in S} P_\mu(s) e^{\mu s}$ on $[0, \infty)$ for distinct rates $\mu \in S$ and nonzero polynomials $P_\mu$ satisfying $\sum_{\mu \in S}((d-1)\deg P_\mu + 1) = N$. The upper bound uses shifted root-of-unity Fourier extraction on truncated exponential polynomials; the lower bound uses multi-mode Alder–Strassen degree peeling.

- **Exact $d$-D real radial rank for all degrees (Theorem 3.5, `separableRank_radial_eq_iff`):** For every $d \geq 2$, separableRank$(x \mapsto \varphi(\|x\|^2)) = N$ over $\mathbb{R}$ if and only if $\varphi$ agrees on $[0, \infty)$ with a well-formed representation whose $d$-dimensional cost (`multiTotalCost`) equals $N$, counting $(d-1)\deg P_i + 1$ for real rates ($\operatorname{Im} \lambda_i = 0$, via real Lagrange interpolation) and $R_i = 2(d-1)\deg P_i + d$ for nonreal rates ($\operatorname{Im} \lambda_i > 0$, via an alternating root-of-unity polynomial-product identity `re_coeff_prod_eq_sum_re` and Alder–Strassen peeling over $\mathbb{R}$). Specializing to $\deg P_i = 0$ gives $r + dc$ ($d$ real terms rather than $2^{d-1}$ per complex Gaussian, `separableRank_radial_gaussian_profile`), and specializing to $d = 2$ recovers totalCost (`separableRank_radial_two_eq_iff`) alongside the minimal differential-annihilator characterization (Theorem 3.6).

- **Comparative formalization of the semigroup step (Section 4):** We compare the 96-line Banach-algebra integral-smoothing proof against a 256-line finite-difference proof in Lean 4 and three finite-dimensional nonstandard-analysis proofs (206 lines in Lean with ultrafilter sequences, 399 lines in Lean with an ultrapower interface, and 275 lines in Isabelle/HOL), all simplified via a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$.

- **Harmonic differential operators, worked examples, and pass-count bounds (Section 5):** Specializing the root-of-unity identity to equal directional-derivative factors recovers and unifies odd and even planar "beach-ball" harmonic operators $\operatorname{Re}((\partial_x + i\partial_y)^m)$ [12, 15], yields exact continuous-domain 2-way separable ranks for analytically differentiated multi-Gaussian steerable filters [11], establishes continuous-domain pass-count lower bounds in 2-D alongside divide-and-conquer pass-tree upper bounds in $d$-D, and compares operation counts and numerical SVD/CP ranks across concrete 2-D and 3-D filter designs.

### 1.2 Related work

**Radial characterizations and Gaussian sum approximations.** Maxwell's kinetic-theory argument (1860) and Schoenberg's characterization of completely monotone radial functions [18] identify single Gaussians and positive Gaussian mixtures among radial kernels. For oscillatory or compactly supported radial kernels in scientific computing, Beylkin and Monzón [2] approximate profiles and Green's functions by short sums of complex exponentials and Gaussians; in image filtering, Lampert and Wirjadi [8] study anisotropic Gaussian separations, while Niemitalo [13, 14] fits circular defocus and low-pass profiles directly with complex-variance Gaussians. None of these works establishes a machine-checked converse for arbitrary signed finite-rank radial kernels or exact $d$-way real CP ranks across higher-degree components.

**Hankel operators, Waring rank, and real vs. complex tensor rank.** Reducing radial kernels via $s_j = x_j^2$ turns $K$ into the $d$-way additive Hankel kernel $\varphi(s_0 + \cdots + s_{d-1})$. In two variables, Kronecker's theorem (1881) relates finite-rank discrete Hankel matrices to rational generating functions and linear recurrences; our continuous half-line version uses translation semigroups on $\mathcal{W}(\varphi)$ to avoid boundary cases and smoothness assumptions beyond continuity. In $d \geq 3$ variables, symmetric and Hankel tensor decompositions go back to Sylvester's apolarity theory (1851) and Comon and Mourrain [3]; Landsberg [9] treats multilinear rank substitution (originating in Alder and Strassen [1]) and the gap between border rank and exact rank, while

Reznick [17] and Qi, Comon, and Lim [16] show that real symmetric/CP rank can strictly exceed complex rank. Our Theorems 3.1 and 3.5 give exact closed-form complex and real $d$-way ranks for the entire continuous Hankel exponential-polynomial class, pinning down the exact real-over-complex gap ($\mathrm{cost}_{d,\mathbb{R}} - \mathrm{cost}_{d,\mathbb{C}} = d - 2$ for each nonreal conjugate pair, independent of polynomial degree!).

**Formalized analysis and nonstandard analysis.** Banach-space calculus, Bochner integration, and normed-algebra exponentials in Lean's Mathlib [10] provide the infrastructure for `Semigroup.lean`, while Isabelle/HOL's multivariate analysis library [6] and Fleuriot's formalization of nonstandard analysis [4] (`HOL-Nonstandard_Analysis`) underpin `Semigroup_NSA.thy`. Section 4 provides a controlled side-by-side comparison of standard and nonstandard formalizations of the same theorem across both systems.

# 2 Definitions, Surjections, and the Two-Dimensional Core

## 2.1 Multi-way separable rank and exponential-polynomial representations

Throughout the paper, $\varphi : \mathbb{R} \to \mathbb{R}$ denotes a real-valued radial profile and $\psi : \mathbb{R} \to \mathbb{C}$ denotes a complex-valued radial profile. Let $\iota$ be a finite index type and $(X_i)_{i\in\iota}$ a family of sets, and let $\mathbb{K}$ be a commutative semiring. In `Defs.lean`:

**Definition 2.1** (`IsSumOfProducts`, `separableRank`)**.** A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of $N$ products* if there exist univariate factors $f : \mathrm{Fin}\, N \to \prod_{i\in\iota}(X_i \to \mathbb{K})$ such that

$$K(x) = \sum_{n\in\mathrm{Fin}\, N} \prod_{i\in\iota} f_n(i, x_i) \qquad \text{for all } x \in \prod_{i\in\iota} X_i.$$

Its *separable rank* is $\mathrm{separableRank}(K) = \inf\{N \in \mathbb{N} : \mathrm{IsSumOfProducts}(K, N)\} \in \mathbb{N} \cup \{\infty\}$ (`ENat`, written `Nat`-with-top in Mathlib).

Separable rank is subadditive under pointwise sums (`IsSumOfProducts.finsetSum`) and monotone under coordinatewise composition (`IsSumOfProducts.comp`). Crucially, coordinatewise *surjections* preserve separable rank exactly (`separableRank_comp_surjective`): if $g_i : Y_i \to X_i$ has a right inverse for each $i \in \iota$, then

$$\mathrm{separableRank}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \mathrm{separableRank}(K). \tag{2}$$

Since $x \mapsto x^2$ maps $\mathbb{R}$ onto $\mathbb{R}_{\geq 0} = [0, \infty)$, applying (2) to $\iota = \mathrm{Fin}\, d$ identifies the $d$-way separable rank of any radial kernel with that of the $d$-way additive Hankel kernel (`separableRank_radial_eq_hankel` in `Translations.lean`):

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \mathrm{separableRank}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto \varphi\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

The same lemma immediately transfers every theorem below to axis-aligned ellipsoidal kernels $\varphi(\sum_j x_j^2/a_j^2)$ ($a_j \neq 0$) and $\ell_p$ superellipsoidal kernels $\varphi(\sum_j |x_j|^p)$ ($p > 0$). Moreover, for $d \geq 2$, padding $(y_0, y_1) \in \mathbb{R}^2$ with $d - 2$ zeros (`padTwo`) yields the slice monotonicity bound (`separableRank_radial_two_le`):

$$\mathrm{separableRank}\big(y \in \mathbb{R}^2 \mapsto \varphi(y_0^2 + y_1^2)\big) \leq \mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

To state real-valued classifications without redundant conjugate duplicates, a *component* $c = (P, \lambda)$ (`Component`) pairs a polynomial $P \in \mathbb{C}[X]$ with a complex rate $\lambda \in \mathbb{C}$ and evaluates on $\mathbb{R}$ as $c.\mathrm{eval}(s) = \mathrm{Re}(P(s)e^{\lambda s})$, with profile $\mathrm{profile}(\mathrm{cs}, s) = \sum_{c\in\mathrm{cs}} c.\mathrm{eval}(s)$.

**Definition 2.2** (`Component.WellFormed`, `IsRepresentation`, `multiTotalCost`)**.** A component $c = (P, \lambda)$ is *well-formed* if $P \neq 0$, $\operatorname{Im}\lambda \geq 0$, and $\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$ (every coefficient of $P$ has zero imaginary part). A list cs is a *representation* (`IsRepresentation`) if all its components are well-formed and their rates $\lambda$ are pairwise distinct. In dimension $d \geq 2$, the $d$-way rendering cost of $c$ (`Component.multiCost`) is

$$c.\mathrm{multiCost}(d) = \begin{cases} (d-1)\deg P + 1 & \text{if } \operatorname{Im}\lambda = 0, \\ 2(d-1)\deg P + d & \text{if } \operatorname{Im}\lambda > 0, \end{cases}$$

with list sum $\mathrm{multiTotalCost}(d, \mathrm{cs}) = \sum_{c \in \mathrm{cs}} c.\mathrm{multiCost}(d)$. At $d = 2$, $c.\mathrm{multiCost}(2) = c.\mathrm{cost}$ ($\deg P + 1$ for real rates, $2(\deg P + 1)$ for nonreal rates, summed in `totalCost`). A representation is *Gaussian* (`IsGaussianRepresentation`) if every $c \in \mathrm{cs}$ has $\deg P = 0$, where $\mathrm{multiTotalCost}(d, \mathrm{cs}) = \mathrm{gaussianTotalCost}(d, \mathrm{cs}) = r + dc$ for $r$ real and $c$ nonreal components.

**Remark 2.3** (Bijection with conjugate-symmetric complex exponential polynomials)**.** Why do the conditions $\operatorname{Im}\lambda \geq 0$ and ($\operatorname{Im}\lambda = 0 \implies P \in \mathbb{R}[X]$) make real representations canonical? Because $\operatorname{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \operatorname{Re}(P(s)e^{\lambda s})$ (where $\overline{P}$ conjugates coefficients) and $\operatorname{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs $(P, \lambda)$ over-parameterize real profiles. Under `Component.WellFormed`, each real rate ($\operatorname{Im}\lambda = 0, P \in \mathbb{R}[X]$) expands in $\mathbb{C}$ to a single term $P(s)e^{\lambda s}$, whereas each upper-half-plane rate ($\operatorname{Im}\lambda > 0$) expands via $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$ (`injective_rates_with_conjugates`). Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$ (`eq_zero_of_sum_expPoly_eq_zero'` in `ExpPoly.lean`), a continuous real profile determines its unordered set of well-formed components uniquely. Notice also that for $\operatorname{Im}\lambda > 0$, the complex rank of this conjugate pair by Theorem 3.1 is $2((d-1)\deg P+1) = 2(d-1)\deg P+2$, so $c.\mathrm{multiCost}(d) - 2((d-1)\deg P+1) = d-2$: every nonreal conjugate pair incurs an exact real-over-complex rank penalty of $d-2$, independent of $\deg P$.

## 2.2 Two-coordinate row span, complexification, and basis-free necessity

When $\iota = \mathrm{Fin}\,2$ and $\mathbb{K}$ is a field, `RowSpan.lean` defines $\mathrm{rowSpan}(K) \leq (X \to \mathbb{K})$ as the $\mathbb{K}$-linear span of the sections $\{x \mapsto K(x, y) : y \in X\}$ (fixing the second coordinate and varying the first) and proves that $\mathrm{separableRank}(K) < \infty$ holds if and only if $\mathrm{rowSpan}(K)$ is finite-dimensional over $\mathbb{K}$, in which case $\mathrm{separableRank}(K) = \mathrm{finrank}_{\mathbb{K}}(\mathrm{rowSpan}\,K)$ (`separableRank_eq_finrank_rowSpan`). For the two-variable Hankel kernel $(s_0, s_1) \mapsto \varphi(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$, the row span is literally the *translate span*

$$\mathcal{W}(\varphi) = \mathrm{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto \varphi(u + t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ (`Complexification.lean`) gives the bound $\mathrm{separableRank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \mathrm{separableRank}_{\mathbb{R}}(K)$ for every finite index set $\iota$, where $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ (`separableRank_ofReal_le`). In *two* coordinates ($\iota = \mathrm{Fin}\,2$), the reverse inequality also holds (`separableRank_ofReal`): $\mathbb{R}$-linearly independent rows in $\mathbb{R}^X$ remain $\mathbb{C}$-linearly independent in $\mathbb{C}^X$ (tested by taking real and imaginary parts of any $\mathbb{C}$-linear relation), so $\mathrm{finrank}_{\mathbb{C}}\,\mathrm{rowSpan}(K_{\mathbb{C}}) = \mathrm{finrank}_{\mathbb{R}}\,\mathrm{rowSpan}(K)$, including when both ranks are $\infty$.

Necessity and the two-dimensional lower bound share a single coordinate-free realization in `TranslationSemigroup.lean`:

1. Suppose $\psi : \mathbb{R} \to \mathbb{C}$ is continuous on $[0, \infty)$ and $\dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty$. Equipping $\mathcal{W}(\psi)$ with a norm transferred across a linear equivalence $\mathcal{W}(\psi) \simeq_{\mathbb{C}} \mathbb{C}^n$ makes $\mathcal{W}(\psi)$ a finite-dimensional Banach space whose induced topology agrees with the subspace topology inherited from the product topology on $\mathbb{R}_{\geq 0} \to \mathbb{C}$ (since all Hausdorff vector topologies on $\mathbb{C}^n$ coincide).

2. Because $\mathcal{W}(\psi)$ is invariant under non-negative shifts $(T_t f)(u) = f(u+t)$ $(t \geq 0)$, continuity of $\psi$ gives pointwise continuity of $t \mapsto T_t f$ on each generator and hence operator-norm continuity of the semigroup $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ on $[0, \infty)$ (`continuous_shiftCLM`).

3. Applying the Banach-algebra semigroup theorem (`Semigroup.lean`, Section 4) yields a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = \mathrm{eval}_0(\exp(tA)\psi_0)$ where $\psi_0 = \psi|_{\mathbb{R}_{\geq 0}} \in \mathcal{W}(\psi)$ (`profile_of_generator`).

4. Decomposing $\psi_0$ into generalized eigenspaces $\ker((A - \mu I)^k)$ of $A$ (`Module.End.iSup_maxGenEigenspace_eq_top` in `Necessity.lean`) truncates the exponential series of $(A - \mu I)$ on each summand, expressing $\psi(t) = \sum_{\mu \in S} Q_\mu(t) e^{\mu t}$ for $t \geq 0$. When $\psi = \varphi_{\mathbb{C}}$ comes from a real profile $\varphi$, averaging $\varphi(t) = \frac{1}{2}(\varphi(t) + \overline{\varphi(t)})$ and pairing $\mu$ with $\bar{\mu}$ using exponential-polynomial uniqueness on $(0, \infty)$ (`ExpPoly.lean`, `RealExpPoly.lean`) regroups the sum into a well-formed real representation (`Representation.lean`).

For the two-dimensional exact rank and annihilator characterization (`Differential.lean`, `ComplexRank.lean`, `Annihilator.lean`), let $D$ denote differentiation within $[0, \infty)$ (right differentiation at 0). Applying Cayley–Hamilton to the generator $A$ on $\mathcal{W}(\psi)$ shows that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ annihilates every orbit $t \mapsto \ell(\exp(tA)v)$, so $\chi_A(D)\psi = 0$ on $[0, \infty)$. Conversely, for $\psi(t) = \sum_i P_i(t) e^{\lambda_i t}$ with distinct rates $\lambda_i$ and $P_i \neq 0$, Hasse-derivative independence shows that $q(D)\psi = 0$ on $[0, \infty)$ if and only if the product annihilator

$$p(X) = \prod_i (X - \lambda_i)^{\deg P_i + 1}$$

divides $q(X)$ (`differentialOperator_sum_expPoly_eq_zero_iff`). Thus $p \mid \chi_A$, giving $\sum_i (\deg P_i + 1) = \deg p \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, while Taylor shift expansion of $P_i(u+t)e^{\lambda_i(u+t)}$ gives the matching upper bound $\dim_{\mathbb{C}} \mathcal{W}(\psi) \leq \sum_i (\deg P_i + 1)$.

# 3 Exact Multi-Way Separable Ranks in $d \geq 2$ Dimensions

In dimension $d \geq 3$, there is no single row-span dimension equal to separableRank$(K)$, and scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ no longer preserves rank. Instead, `MultiComplexRank.lean` and `MultiRealRank.lean` establish exact $d$-way upper bounds by root-of-unity and Lagrange identities and matching lower bounds by mapping functional decompositions onto finite-dimensional multilinear forms.

## 3.1 Bridging function spaces and multilinear tensor rank

Suppose $V$ is a $\mathbb{K}$-vector space over a field $\mathbb{K}$, $v : \alpha \to V$ is an evaluation map whose image spans $V$ ($\mathrm{span}_{\mathbb{K}}(\mathrm{range}\, v) = \top$), and $M : V^d \to \mathbb{K}$ is a $d$-multilinear map (`MultilinearMap` $\mathbb{K}$ (Fin $d \to V$) $\mathbb{K}$). Lemma `exists_linear_factors_of_span_eq_top` proves that if the composed kernel $s \in \alpha^d \mapsto M(v(s_0), \ldots, v(s_{d-1}))$ admits a functional decomposition (1) of rank $R$ with arbitrary set-theoretic factors $f_{n,j} : \alpha \to \mathbb{K}$, then there exist *linear* functionals $\ell_{n,j} \in V^*$ such that

$$M(u_0, \ldots, u_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u_j) \qquad \text{for all } u_0, \ldots, u_{d-1} \in V. \tag{5}$$

Indeed, selecting a $\mathbb{K}$-basis $\{v(s_b)\}_{b \in B}$ of $V$ from range$(v)$ defines unique linear functionals $\ell_{n,j} \in V^*$ satisfying $\ell_{n,j}(v(s_b)) = f_{n,j}(s_b)$; both sides of (5) are $d$-multilinear maps that agree on every $d$-tuple of basis vectors, so they agree on $V^d$. To verify $\mathrm{span}_{\mathbb{K}}(\mathrm{range}\, v) = \top$ when $\dim_{\mathbb{K}} V < \infty$, Lemma `span_range_eq_top_of_dual` shows by quotient duality that it suffices to check $L \circ v = 0 \implies L = 0$ for every $L \in V^*$.

### 3.2 Exact $d$-way complex exponential-polynomial rank

Fix $d \geq 2$, a finite index set $\iota$, injective rates $\lambda : \iota \to \mathbb{C}$, and nonzero polynomials $P_i \in \mathbb{C}[X]$ of degree $k_i = \deg P_i$ $(i \in \iota)$.

**Upper bound via shifted root-of-unity interpolation.** For a single polynomial $P \in \mathbb{C}[X]$ of degree $k = \deg P$, put $L = (d-1)k + 1 \geq 1$ and $\zeta = \exp(2\pi i/L)$. Using the $q$-th Hasse derivative $D^{[q]}P$ (so $P(s_0 + t) = \sum_{q=0}^{k} (D^{[q]}P)(s_0)\, t^q$), define the reversed Hasse polynomial on coordinate 0 and the truncated exponential polynomials on coordinates $j \in \{1, \ldots, d-1\}$ (`revHassePoly`, `truncExpPoly`):

$$R(s_0; X) = \sum_{q=0}^{k} q!\, (D^{[q]}P)(s_0)\, X^{k-q}, \qquad E_k(s_j; X) = \sum_{q=0}^{k} \frac{s_j^q}{q!} X^q,$$

$$Q(X) = R(s_0; X) \prod_{j=1}^{d-1} E_k(s_j; X).$$

Binomial convolution gives $[X^q] \prod_{j=1}^{d-1} E_k(s_j; X) = (s_1 + \cdots + s_{d-1})^q / q!$ for all $0 \leq q \leq k$, so the coefficient of $X^k$ in $Q(X)$ is precisely $P(s_0 + s_1 + \cdots + s_{d-1})$. Crucially, because $d \geq 2$, we have $k < L$ and $\deg Q \leq k + (d-1)k = k + L - 1 < k + L$. For *any* polynomial $Q$ with $\deg Q < k + L$ and $k < L$, discrete Fourier averaging over only $L$ roots of unity extracts $[X^k]Q(X)$ (`coeff_eq_sum_rootsOfUnity`):

$$[X^k]Q(X) = \frac{1}{L} \sum_{r=0}^{L-1} \zeta^{-rk}\, Q(\zeta^r),$$

because every monomial index $n \in \{0, \ldots, k+L-1\}$ satisfies $-L < -k \leq n-k < L$, so $L \mid (n-k)$ holds if and only if $n = k$. Multiplying the $j$-th factor of $Q(\zeta^r)$ by $e^{\lambda s_j}$ (and absorbing $\frac{1}{L}\zeta^{-rk}$ into coordinate 0) expresses $P(\sum_{j=0}^{d-1} s_j) e^{\lambda \sum_{j=0}^{d-1} s_j}$ as a sum of $L = (d-1)\deg P + 1$ coordinatewise products (`isSumOfProducts_expPoly`).

**Lower bound via Alder–Strassen degree peeling.** Let $m_0(i) = \deg P_i + 1$ and form the coefficient evaluation space $V = \prod_{i \in \iota} \mathbb{C}^{m_0(i)}$ with $v(s)_{i,q} = \frac{s^q}{q!} e^{\lambda_i s}$ $(0 \leq q < m_0(i))$. By univariate exponential-polynomial uniqueness (`eq_zero_of_sum_expPoly_eq_zero'`), no nonzero functional in $V^*$ vanishes on range$(v)$, so $\mathrm{span}_{\mathbb{C}}(\mathrm{range}\, v) = \top$ (`span_expMonomial_eq_top`). Under $v$, the Hankel kernel corresponds to the $d$-multilinear truncated-convolution form (`truncForm`) with cutoffs $c_i \leq m_0(i)$,

$$\mathcal{T}_c(u^{(0)}, \ldots, u^{(d-1)}) = \sum_{i \in \iota} \sum_{q_0 + \cdots + q_{d-1} < c_i} w_i(q_0 + \cdots + q_{d-1}) \prod_{j=0}^{d-1} u^{(j)}_{i,q_j},$$

where $w_i(c_i - 1) \neq 0$ whenever $c_i > 0$. Lemma `truncForm_rank_ge` proves

$$\sum_{i:\, c_i > 0} \big((d-1)(c_i - 1) + 1\big) \leq R$$

for every rank-$R$ decomposition (5) by strong induction on $S = \sum_{i \in \iota} c_i$:

- *Case $c_{i_0} = 1$:* Evaluating $\mathcal{T}_c$ at $(e_{i_0,0}, \ldots, e_{i_0,0})$ gives $w_{i_0}(0) \neq 0$, selecting an index $n_0$ with $\ell_{n_0,0}(e_{i_0,0}) \neq 0$. Let $\pi_{\neq i_0} : V \to V$ zero out block $i_0$. Substituting $\varphi_0(x) = \pi_{\neq i_0}(x) - \frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 and $\pi_{\neq i_0}$ on modes $j \geq 1$ annihilates $\ell_{n_0,0} \circ \varphi_0$ while reducing $c_{i_0}$ from 1 to 0 (since $d \geq 2$ zeroes block $i_0$ on mode 1), leaving at most $R - 1$ terms.

- *Case $c_{i_0} \geq 2$:* Set $s_{\text{top}} = e_{i_0,c_{i_0}-1}$ and $s_0 = e_{i_0,0}$ ($s_{\text{top}} \neq s_0$). Because $(c_{i_0}-1)+(c_{i_0}-1) \geq c_{i_0}$, any rank-1 perturbation of the form $\psi(x) = x - \gamma(x)s_{\text{top}}$ leaves $\mathcal{T}_c$ unchanged whenever another coordinate equals $s_{\text{top}}$. Stepping through modes $j = 0, \ldots, d-2$ sequentially evaluates $\mathcal{T}_c$ with $u^{(j)} = s_{\text{top}}$ and $u^{(k)} = \psi_k(s_0)$ ($k \neq j$) to find $d-1$ distinct indices $D \subseteq \operatorname{Fin} R$ ($|D| = d-1$) killed on modes $0, \ldots, d-2$ by rank-1 perturbations $\psi_j$ along $s_{\text{top}}$. On modes $0, \ldots, d-2$ we then truncate block $i_0$ to indices $q < c_{i_0} - 1$ before applying $\psi_j$, while on mode $d-1$ we shift block $i_0$'s degree indices up by $+1$ ($e_{i_0,q} \mapsto e_{i_0,q+1}$ for $q < c_{i_0} - 1$): this transforms $\mathcal{T}_c$ into the reduced form $\mathcal{T}_{c'}$ ($c' = c[i_0 \mapsto c_{i_0} - 1]$) with shifted weights $w'_{i_0}(m) = w_{i_0}(m+1)$—preserving the nonzero top weight $w'_{i_0}(c'_{i_0} - 1) = w_{i_0}(c_{i_0} - 1) \neq 0$—while dropping all $d-1$ terms in $D$.

Combining this lower bound at $c = m_0$ with the upper bound and Section 2 gives:

**Theorem 3.1** ($d$-D Complex Radial Rank, `separableRank_radial_complex_eq_iff`)**.** *Let $d \geq 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = N$$

*if and only if there exist a finite set $S \subset \mathbb{C}$ and polynomials $(P_\mu)_{\mu \in S}$ with $P_\mu \neq 0$ such that $\psi(s) = \sum_{\mu \in S} P_\mu(s) e^{\mu s}$ for all $s \geq 0$ and $\sum_{\mu \in S} \big((d-1)\deg P_\mu + 1\big) = N$.*

### 3.3 Exact $d$-way real rank for arbitrary degrees (`MultiRealRank.lean`)

Before treating polynomial degrees, note that for constant factors ($d \geq 1$), expanding $\operatorname{Re}(\prod_{j=0}^{d-1} z_j)$ into real and imaginary parts gives $2^{d-1}$ real products, whereas $d$ real products always suffice:

**Proposition 3.2** (Alternating $2d$-th root-of-unity identity, `re_prod_eq_sum_re`)**.** *For every integer $d \geq 1$ and all $z_0, \ldots, z_{d-1} \in \mathbb{C}$,*

$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{m=0}^{d-1} (-1)^m \prod_{j=0}^{d-1} \operatorname{Re}\left(z_j e^{-i\pi m/d}\right). \tag{6}$$

*Proof.* Write $\operatorname{Re}(z_j e^{-i\pi m/d}) = \frac{1}{2}\left(\overline{z_j} e^{i\pi m/d} + z_j e^{-i\pi m/d}\right)$ and expand over subsets $S \subseteq \{0, \ldots, d-1\}$: multiplying by $(-1)^m = e^{i\pi m}$ yields the geometric progression $w_S^m$ with $w_S = e^{2\pi i|S|/d}$ ($w_S^d = 1$), whose sum over $0 \leq m < d$ equals $d$ for $S \in \{\emptyset, \{0, \ldots, d-1\}\}$ and 0 for $0 < |S| < d$. □

Absorbing $\frac{2^{d-1}}{d}(-1)^m$ into coordinate 0 shows that the real part of a rank-$N$ complex $d$-way kernel has real rank at most $Nd$ (`IsSumOfProducts.re`), establishing qualitative $d$-D classification:

**Theorem 3.3** ($d$-D Finite-Rank Classification, `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. Then*

$$\begin{aligned} &\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) < \infty \\ &\iff \exists\, \mathrm{cs},\ \operatorname{IsRepresentation}(\mathrm{cs}) \wedge \forall s \geq 0,\ \varphi(s) = \operatorname{profile}(\mathrm{cs}, s). \end{aligned}$$

For degree $k = \deg P \geq 1$, however, multiplying Theorem 3.1's complex rank $(d-1)k+1$ by $d$ via `IsSumOfProducts.re` gives $d(d-1)k + d$, which is non-optimal. Instead, sharp upper bounds differ for real and nonreal rates (`isSumOfProducts_radial_multiComponent`):

- *Real rates ($\operatorname{Im}\lambda = 0$, $P \in \mathbb{R}[X]$ of degree $k$):* On coordinates $1, \dots, d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1} E_k(x_j^2; X) \in \mathbb{R}[X]$ has $\deg A \le (d-1)k < L := (d-1)k+1$. Real Lagrange interpolation (`Lagrange.eq_interpolate`) at the $L$ real nodes $0, 1, \dots, L-1 \in \mathbb{R}$ expresses each $[X^q]A(X) = (\sum_{j=1}^{d-1} x_j^2)^q/q!$ $(0 \le q \le k)$ as an $\mathbb{R}$-linear combination of $\prod_{j=1}^{d-1} E_k(x_j^2; r)$, yielding $(d-1)k+1$ real $d$-way products.
- *Nonreal rates ($\operatorname{Im}\lambda > 0$, $P \in \mathbb{C}[X]$ of degree $k$):* We generalize Proposition 3.2 from scalars to polynomials of degree at most $k$ $(d \ge 2)$:

**Proposition 3.4** (Universal root-of-unity polynomial-product identity, `re_coeff_prod_eq_sum_re`)**.** *Let $d \ge 2$, $k \ge 0$, and $R = 2(d-1)k + d$. For any $U_0, \dots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \le k$,*

$$\operatorname{Re}\left( [X^k] \prod_{j=0}^{d-1} U_j(X) \right) = \frac{2^{d-1}}{R} \sum_{r=0}^{R-1} (-1)^r \prod_{j=0}^{d-1} \operatorname{Re}\Big( U_j\Big( e^{\frac{2\pi i r}{R}} \Big)\, e^{-\frac{i\pi r(2k+1)}{R}} \Big). \tag{7}$$

*Proof.* Expanding $\operatorname{Re} w = \frac{1}{2}(\bar{w}+w)$ over subsets $S \subseteq \{0, \dots, d-1\}$ of size $s = |S|$ and coefficient indices $q \in \{0, \dots, k\}^d$, and noting that $(-1)^r = e^{i\pi r R/R}$ with $R = (2k+1)d - 2k$, gives the $r$-th phase $\exp(2\pi i r K(S,q)/R)$ where

$$K(S,q) = (2k+1)s + \sum_{j \notin S} q_j - \sum_{j \in S} q_j - k \in \mathbb{Z}.$$

Because $0 \le q_j \le k$ and $d \ge 2$: (i) when $S = \emptyset$ $(s = 0)$, $-R < -k \le K(\emptyset, q) = \sum_j q_j - k \le (d-1)k < R$, so $R \mid K(\emptyset, q) \iff \sum_j q_j = k$; (ii) when $S = \{0, \dots, d-1\}$ $(s = d)$, $K = R + (k - \sum_j q_j)$, so $R \mid K \iff \sum_j q_j = k$; and (iii) when $1 \le s \le d-1$, $K(S,q) \ge (2k+1)s - sk - k = (k+1)s - k \ge 1$ and $K(S,q) \le (2k+1)s + (d-s)k - k = (k+1)s + (d-1)k \le 2(d-1)k + d - 1 = R - 1$, so $R \nmid K(S,q)$ and the geometric sum over $0 \le r < R$ vanishes identically. Applying (7) to $U_0(X) = e^{\lambda x_0^2} R(x_0^2; X)$ and $U_j(X) = e^{\lambda x_j^2} E_k(x_j^2; X)$ $(j \ge 1)$ yields $2(d-1)k + d$ real $d$-way products. □

**Unified Alder–Strassen lower bound over $\mathbb{R}$ (`realExpForm_rank_ge`).** Let $G = (\prod_{i \in \iota_{\mathbb{R}}} \mathbb{R}^{m_{\mathbb{R}}(i)}) \times (\prod_{k \in \iota_{\mathbb{C}}} \mathbb{C}^{m_{\mathbb{C}}(k)})$ with $m(i) = \deg P_i + 1$. Conjugate-rate uniqueness shows via duality that the exponential-monomial curve $s \mapsto ((s^q/q!\, e^{\mu_i s}), (s^q/q!\, e^{\lambda_k s}))$ spans $G$ over $\mathbb{R}$ (`span_realExpEval_eq_top`), transferring any rank-$R$ decomposition of the Hankel kernel to $\mathbb{R}$-linear functionals $\ell_{n,j} \in G^*$ for the combined truncated-convolution form `realExpForm`. Note first that every $L \in \operatorname{Hom}_{\mathbb{R}}(\mathbb{C}, \mathbb{R})$ has a nonzero kernel vector $w \in \ker L \setminus \{0\}$ (`exists_ker_ne_zero`), from which induction on $d \ge 2$ gives $d \le R$ for any single nonzero complex product $\operatorname{Re}(\beta \prod_{j=0}^{d-1} z_j)$ (`single_complex_rank_ge`). Strong induction on the cutoff sum then peels:

1. *Real cutoffs (`realExpForm_rank_ge`):* Identical to Section 3.2 (1 term when $c_{\mathbb{R}}(i_0) = 1$, $d-1$ terms when $c_{\mathbb{R}}(i_0) \ge 2$), while keeping the $\mathbb{C}$-coordinates of $G$ fixed so all nonreal blocks are untouched.
2. *Nonreal cutoff $c_{\mathbb{C}}(k_0) = 1$ (`cplxTruncForm_rank_ge`):* On mode 0, evaluating at $1_{k_0}$ and at $w_1 \in \ker(\ell_{n_1,0}|_{\mathbb{C}e_{k_0,0}}) \setminus \{0\}$ kills 2 distinct terms, and on each mode $j = 1, \dots, d-2$, `single_complex_rank_ge` forces one additional active term outside the killed set $D$ (else $|D| = j+1 < d$ terms would represent $\operatorname{Re}(\beta_{k_0} \prod z_q)$), reaching $|D| = d$ before zeroing block $k_0$ on mode $d-1$.
3. *Nonreal cutoff $c_{\mathbb{C}}(k_0) \ge 2$:* Let $q_{\text{top}} = c_{\mathbb{C}}(k_0) - 1 \ge 1$ with weight $\beta_{\text{top}} \ne 0$. In each mode $j = 0, \dots, d-2$, evaluating at $u(z)$ with $\text{emb}_{\text{top}}(z)$ on mode $j$, $\text{emb}_0(\overline{\beta_{\text{top}} z})$ on mode $d-1$,

and $\psi_m(\mathrm{emb}_0(1))$ elsewhere gives $|\beta_{\mathrm{top}} z|^2 > 0$ for every $z \in \mathbb{C} \setminus \{0\}$ (since $2q_{\mathrm{top}} \geq c_{\mathbb{C}}(k_0)$ kills all top-degree cross-terms). Testing $z = 1$ and $z = w_1 \in \ker(\ell_{n_1,j} \circ \mathrm{emb}_{\mathrm{top}}) \setminus \{0\}$ kills 2 distinct terms per mode—$2(d-1)$ terms across modes $0, \ldots, d-2$—after which low-degree truncation on modes $0, \ldots, d-2$ and $+1$ index shifting on mode $d-1$ reduce $c_{\mathbb{C}}(k_0) \mapsto c_{\mathbb{C}}(k_0) - 1$ with shifted top weight $\beta_{\mathrm{top}} \neq 0$.

**Theorem 3.5** (Exact $d$-D Real Radial Rank, `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\mathrm{separableRank}\left( x \in \mathbb{R}^d \mapsto \varphi\left( \sum_{j=0}^{d-1} x_j^2 \right) \right) = N$$

$$\iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge (\forall s \geq 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s)) \wedge \mathrm{multiTotalCost}(d, \mathrm{cs}) = N.$$

*In particular, for a Gaussian representation with $r$ real and $c$ nonreal rates,* $\mathrm{separableRank} = r + dc$ *(`separableRank_radial_gaussian_profile`), and in $d = 2$,* $\mathrm{separableRank} = \mathrm{totalCost}(\mathrm{cs})$ *(`separableRank_radial_two_eq_iff`).*

**Theorem 3.6** (Minimal Differential Annihilator)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with finite-dimensional translate span and two-dimensional radial rank* $\mathrm{separableRank}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = N < \infty$ *(`exists_minimal_differential_annihilator`). There exists a monic polynomial $p \in \mathbb{C}[X]$ of degree $N$ such that for every $q \in \mathbb{C}[X]$, the half-line differential equation $q(D)\varphi_{\mathbb{C}} = 0$ holds on $[0, \infty)$ (where $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$) if and only if $p \mid q$.*

**Worked example ($d = 3, \deg P = 1$): exact rank vs. complex rank and border rank.** Consider $K(x_0, x_1, x_2) = (x_0^2 + x_1^2 + x_2^2)\cos(x_0^2 + x_1^2 + x_2^2)$ on $\mathbb{R}^3$, a single well-formed component $P(s) = s$, $\lambda = i$ ($\deg P = 1, \mathrm{Im}\,\lambda = 1 > 0$):

- *Complex rank $= 6$:* Over $\mathbb{C}$, $K_{\mathbb{C}}$ splits into $\frac{1}{2} s e^{is} + \frac{1}{2} s e^{-is}$, giving exact complex 3-way rank $((3-1) \cdot 1 + 1) + ((3-1) \cdot 1 + 1) = 6$ by Theorem 3.1.
- *Pointwise limit of rank-6 kernels (border rank $\leq 6$):* As an unformalized analytical contrast, for every $h > 0$ the difference quotient $\frac{1}{h}\left[\mathrm{Re} \prod_{j=0}^{2} (1 + h x_j^2) e^{i x_j^2} - \mathrm{Re} \prod_{j=0}^{2} e^{i x_j^2}\right]$ converges pointwise (and uniformly on compact sets) to $K(x)$ as $h \to 0$, and by Proposition 3.2 each $\mathrm{Re}(\prod_{j=0}^{2} \cdots)$ uses 3 real 3-way products, totaling $3 + 3 = 6$ real products at every $h > 0$ with diverging $O(h^{-1})$ weights.
- *Exact real rank $= 7$:* By Theorem 3.5, $\mathrm{separableRank}_{\mathbb{R}}(K) = 2(3-1) \cdot 1 + 3 = 7$ (achieved with bounded weights by Proposition 3.4 at $R = 7$, vs. 9 from naive `IsSumOfProducts.re`). To see why 6 terms cannot represent $K$ exactly, pass to $W = \mathbb{C}[T]/(T^2) \cong \mathbb{C} \oplus \mathbb{C}\varepsilon$ ($\varepsilon^2 = 0$) where $\mathcal{M}(u, v, w) = \mathrm{Re}(u_1 v_0 w_0 + u_0 v_1 w_0 + u_0 v_0 w_1)$. On $E = 0 \oplus \mathbb{C}\varepsilon \cong \mathbb{R}^2$, the slice $\mathcal{M}((0, z_1), (v_0, 0), (w_0, 0)) = \mathrm{Re}(z_1 v_0 w_0)$ forces the mode-0 functionals $\{\ell_n|_E\}$ to span $E^*$ ($\dim_{\mathbb{R}} E^* = 2$); choosing a basis $\{\ell_0|_E, \ell_1|_E\}$ defines $\psi_0(u_0) = (u_0, c_0(u_0))$ killing $n = 0, 1$. Because $\varepsilon^2 = 0$, $\mathcal{M}(\psi_0(u_0), (0, v_1), (w_0, 0)) = \mathrm{Re}(u_0 v_1 w_0)$ is unchanged by $c_0$, so the remaining functionals $\{m_n|_E : n \geq 2\}$ on mode 1 again span $E^*$, killing two more terms $n = 2, 3$ via $\psi_1(v_0) = (v_0, c_1(v_0))$. Finally, $\mathcal{M}(\psi_0(u_0), \psi_1(v_0), (0, w_1)) = \mathrm{Re}(u_0 v_0 w_1)$ must be represented by the remaining $R - 4$ terms, and `single_complex_rank_ge` gives $R - 4 \geq 3$, hence $R \geq 7$.

## 4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s + t) = M(s)M(t)$ for all $s, t \geq 0$, and $M$ is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \geq 0$.

### 4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

**Integral smoothing (`Semigroup.lean`).** Since $M(0) = 1$ and $M$ is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\,du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making $I_\delta$ invertible in $\mathcal{A}$ by Neumann series. For $t \geq 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\,du$ shows by the fundamental theorem of calculus that $M$ is right-differentiable on $[0,\infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

**Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`).** Without integration, if $\|M(u) - 1\| \leq \frac{1}{2}$ on $[0,\delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_k) - 1)/h_k \to A$; along $h_k$, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

**Nonstandard analysis with homogeneous common scale.** Euler's heuristic sets $A = \mathrm{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$. Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (which is infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{8}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0,1]$ and $\left\|\frac{x}{\eta}\right\| \leq 1$, both rescaled quantities are bounded by 1 *without any analytical estimate*. In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \mathrm{st}(x/\eta)$ with $\|B\| \leq 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0,1]$. Now compare $N = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N\varepsilon \approx sa$) against $N$ steps of $\exp(\eta B)$ (arriving at time $N\eta \approx s$) for an arbitrary standard $s \geq 0$:

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \leq \left\|\frac{x}{\eta} - B\right\| + \left\|B - \frac{\exp(\eta B) - 1}{\eta}\right\| \approx 0.$$

Summing $N$ single-step errors of size $\eta \cdot o(1)$ with $N\eta \leq s$ yields $M(sa) = \exp(sB)$ for all standard $s \geq 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \geq 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$! Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

### 4.2 Three nonstandard formalizations and quantitative comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

1. **Isabelle/HOL:** `Isabelle/Semigroup_NSA.thy` uses `HOL-Nonstandard_Analysis`, which provides hyperreals, hypernaturals, star extensions `*f*`, and the `transfer` proof method. Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \leq e^h$ with $h \geq 0$, then $\|u^n - v^n\| \leq ne^{nh}\|u - v\|$, proved by induction from $u^{n+1} - v^{n+1} = u(u^n - v^n) + (u - v)v^n$. Since $\|M(\varepsilon)\| \leq 1 + \|x\| \leq 1 + \eta \leq e^\eta$ and $\|\exp(\eta B)\| \leq e^{\eta\|B\|} \leq e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count $N$. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps using compact-interval bounds via `norm_replace_steps_le` instead.)

Table 1: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$. Every project helper lemma in each proof's module dependencies is counted in full (excluding standard Mathlib and Isabelle/HOL library imports).

| Proof method | Prover / files | Phys. | Nonbl. | Scope |
|---|---|---|---|---|
| Integral smoothing (main) | Lean 4 (`Semigroup.lean`) | 96 | 87 | Any Banach algebra |
| Finite differences | Lean 4 (`GeneratorBound`, `FD`) | 256 | 233 | Finite-dim. |
| NSA (ultrafilter seq.) | Lean 4 (`NonstandardSemigroup`) | 206 | 190 | Finite-dim. |
| NSA (ultrapower API) | Lean 4 (`Ultrapower*` + shared) | 399 | 344 | Finite-dim. |
| NSA (star + `transfer`) | Isabelle (`Semigroup_NSA.thy`) | 275 | 253 | Finite-dim. ($\|1\| = 1$) |

2. **Lean 4 with ultrafilter sequences:** `Experiments/NonstandardSemigroup.lean` expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining $B$ in one line via `IsCompact.ultrafilter_le_nhds`.

3. **Lean 4 in nonstandard vocabulary:** In `Experiments/`, `Ultrapower.lean` and `UltrapowerSemigroup.lean` build a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\mathrm{h}}$, one-variable transfer lemmas) and write Euler's proof in classical nonstandard notation.

Table 1 summarizes the exact line counts. These numbers reflect *library amortization* rather than intrinsic proof complexity: `Semigroup.lean` (96 lines) is shortest and holds in every Banach algebra because Mathlib amortizes thousands of lines of Bochner integration, Neumann series, and the fundamental theorem of calculus behind compact API lemmas, whereas compactness-based proofs are inherently finite-dimensional. Across different proof styles within Lean, the 2× gap between integral smoothing (96 lines) and ultrafilter sequences (206 lines) remained stable even after eliminating the 124-line difference-quotient bound via homogeneous coordinates (8). Moreover, discounting the 195-line interface `Ultrapower.lean` as reusable library code leaves $169 + 35 = 204$ lines: without automated transfer, nonstandard vocabulary in Lean merely wraps filter limits. In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, but 59 lines of local library gaps (Heine–Borel vector standard parts and half-line continuity via $M(\max(t, 0))$) plus stating transferred formulas at both standard and nonstandard types keep the total at 275 lines.

# 5 Applications to Isotropic and Steerable Filtering and 3-D Propagation

## 5.1 Harmonic "beach-ball" differential operators and steerable kernels

Specializing Proposition 3.2 to equal symbolic factors $z_0 = \cdots = z_{d-1} = \partial_x + i\partial_y$ (writing $m$ for the harmonic order and $D_\alpha = \cos\alpha\,\partial_x + \sin\alpha\,\partial_y$ for the directional derivative along angle $\alpha$) gives, for *every* integer $m \geq 1$ (both odd and even),

$$\mathrm{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m} \sum_{k=0}^{m-1} (-1)^k D^m_{\pi k/m}. \tag{9}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ ($n = 0, \ldots, 2m-1$) of radius $h \to 0$, antipodal points satisfy $D_{\theta_{k+m}} = -D_{\theta_k}$ while Fourier orthogonality of $(-1)^k = e^{i\pi k}$ annihilates all Taylor orders below $m$. For $f \in C^m$ near $(x, y)$, (9) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order ($m = 2N + 1$)

hexagonal/decagonal "beach-ball" operators of Niemitalo [12] and the even-order operators of Perona [15] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x+h\cos\frac{\pi n}{m},\, y+h\sin\frac{\pi n}{m}\Big)=\frac{1}{(m-1)!\,2^{m-2}}\operatorname{Re}\big((\partial_x+i\partial_y)^m\big)f(x,y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x+i\partial_y)^m)$. Recognizing $\partial_{\bar z}=\frac{1}{2}(\partial_x+i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial_{\bar z}^m$ to *any* smooth radial kernel $\varphi(x^2+y^2)=\varphi(z\bar z)$ produces no lower-order radial derivatives: because $\partial_{\bar z}(z)=0$, induction gives

$$\left(\frac{\partial_x+i\partial_y}{2}\right)^m\varphi(x^2+y^2)=(x+iy)^m\,\varphi^{(m)}(x^2+y^2)=r^m e^{im\theta}\,\varphi^{(m)}(r^2). \tag{10}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with $r$ real and $c$ nonreal upper-half-plane rates (all $\lambda_i\neq 0$, approximating, e.g., an ideal circular low-pass filter [13]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by $\lambda_i^m$) of continuous 2-way rank $R=r+2c$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [11]—absorbs the monomials of $\operatorname{Re}((x+iy)^m)$ and $\operatorname{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2\rfloor+1\big)(r+2c)\quad\text{for }\cos(m\theta),\qquad\lceil m/2\rceil(r+2c)\quad\text{for }\sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms (grid aliasing at special rates such as $\operatorname{Im}\lambda_i\in\pi\mathbb{Z}$ can collapse the sampled discrete rank further).

**Continuous-domain pass-count bounds for simultaneous steerable pairs.** In applications such as orientation estimation [11], one evaluates both continuous-domain convolutions $(C_m * I, S_m * I)$ of a harmonic pair simultaneously. Can sharing intermediate 1-D continuous convolutions reduce the pass count below separate evaluation $2(\lfloor m/2\rfloor+1)R+2\lceil m/2\rceil R=2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m,S_m)=(A_c+B_c,A_s+B_s)$ into $P_y$ shared vertical passes followed by $\operatorname{rank}(A_c)+\operatorname{rank}(A_s)$ horizontal passes plus $P_x$ shared horizontal passes followed by $\operatorname{rank}(B_c)+\operatorname{rank}(B_s)$ vertical passes, then

$$\begin{aligned}\dim\operatorname{rowSpan}_x(C_m,S_m)&\le\operatorname{rank}(A_c)+\operatorname{rank}(A_s)+P_x,\\ \dim\operatorname{colSpan}_y(C_m,S_m)&\le P_y+\operatorname{rank}(B_c)+\operatorname{rank}(B_s).\end{aligned}$$

Because $C_m$ and $S_m$ on $\mathbb{R}^2$ span all powers $\{x^0,\dots,x^m\}g_n(x)$ in $x$ and $\{y^0,\dots,y^m\}g_n(y)$ in $y$ (for $m=1$, $h_x$ is odd-in-$x$/even-in-$y$ while $h_y$ is even-in-$x$/odd-in-$y$), adding the two inequalities proves that *every* two-stage continuous 1-D schedule requires at least $(m+1)R+(m+1)R=2(m+1)R$ passes ($4R$ passes for $(h_x,h_y)$). Savings instead arise in three ways: (i) definite even/odd parity of every 1-D factor $u^k g_n(u)$ halves the multiplications per sampled pass via $I[y+\ell]\pm I[y-\ell]$ folding while fusing input loads; (ii) across *multiple* harmonic orders $m=0,\dots,M$, vertical passes $\{I *_y (y^k g_n)\}_{k=0}^{M}$ are shared linearly ($5R$ passes instead of $6R$ for low-pass plus gradient; $10R$ instead of $12R$ for $m=1$ plus $m=3$); and (iii) for the $d$-dimensional gradient $(h_{x_0},\dots,h_{x_{d-1}})$ of a multi-Gaussian radial kernel of $d$-way rank $R_d=r+dc$ ($d\ge 3$), recursively splitting $\{0,\dots,d-1\}$ into halves of sizes $\lfloor d/2\rfloor$ and $\lceil d/2\rceil$ and pre-convolving each half with the undifferentiated factors reduces $d^2R_d$ passes to $T(d)R_d$ with $T(1)=1$ and $T(d)=T(\lfloor d/2\rfloor)+T(\lceil d/2\rceil)+d\le d(\lceil\log_2 d\rceil+1)$ ($8R_3$ vs. $9R_3$ in $d=3$; $12R_4$ vs. $16R_4$ in $d=4$).

Table 2: Comparison of exact continuous-domain real separable rank $R$, 1-D passes, and scalar multiply–adds per output sample ($dRL$) against direct spatial convolution ($L^d$) and naive real-part expansion at tap widths $L = 41$ and $L = 161$ (from [11]).

| Kernel family | Params | Direct (41/161) | Naive | Exact $R$ | Passes | Ops (41/161) |
|---|---|---|---|---|---|---|
| 2-D circular low-pass $W$ | $d$=2, $c$=2 | 1,681 / 25,921 | 8 | 4 | 8 | 328 / 1,288 |
| 2-D gradient $(h_x, h_y)$ | $d$=2, $m$=1, $c$=2 | 3,362 / 51,842 | 16 | 4+4 | 16 | 656 / 2,576 |
| 2-D beach ball $(C_3, S_3)$ | $d$=2, $m$=3, $c$=2 | 3,362 / 51,842 | 32 | 8+8 | 32 (40*) | 1,312 / 5,152 |
| 3-D Gaussian prop. | $d$=3, $r$=1, $c$=2 | 68,921 / $4.17\times10^6$ | 27 | 7 | 21 | 861 / 3,381 |
| 3-D osc. shell (deg =1) | $d$=3, deg =1, $c$=1 | 68,921 / $4.17\times10^6$ | 27 | 7 | 21 | 861 / 3,381 |

Table 3: Structure and source line counts of the `KernelRankLean` library.

| Module(s) | Mathematical content | Phys. | Nonbl. |
|---|---|---|---|
| `Defs`, `KernelRankLean` | Multi-way rank, surjections, slices, costs | 213 | 180 |
| `RowSpan`, `Complexification` | 2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance | 221 | 189 |
| `Translations` | $d$-D radial-to-Hankel reduction; `translateSpan` | 92 | 80 |
| `ExpPoly`, `RealExpPoly` | Exponential-polynomial uniqueness | 198 | 170 |
| `Semigroup`, `TranslationSemigroup` | Banach semigroups; shift semigroup | 262 | 219 |
| `Necessity`, `Representation` | Generalized eigenvectors; conjugate pairing | 222 | 198 |
| `Differential`, `ComplexRank` | Half-line operators $q(D)$; 2-D complex rank | 318 | 284 |
| `Annihilator` | Minimal constant-coefficient annihilator | 69 | 62 |
| `MultiComplexRank` | Exact $d$-way complex rank $\sum_i((d-1)\deg P_i + 1)$ | 1,015 | 978 |
| `MultiRealRank` | Exact $d$-way real rank (`multiTotalCost`) | 2,562 | 2,521 |
| `Main` | Unified $d$-D and 2-D main theorems | 152 | 140 |
| **Total (`KernelRankLean`)** | | **5,324** | **5,021** |

### 5.2 Three-dimensional filtering, wave propagation, and concrete operation counts

In three dimensions ($d = 3$), direct spatial convolution with an $L \times L \times L$ isotropic kernel—such as a cryo-EM/tomography bandpass filter or a coarse-grid dispersion-free time-step propagator for the band-limited constant-speed acoustic wave equation, whose spatial-frequency symbol depends radially on $\|\xi\|^2$—scales as $L^3$ per voxel. Approximating the radial profile by $r$ real and $c$ complex-conjugate Gaussians yields exact real 3-way separable rank $r + 3c$ by Theorem 3.5: Proposition 3.2 realizes each complex Gaussian in 3 real 3-way terms (9 real 1-D passes on real-valued volume buffers, vs. $2^{3-1} = 4$ terms / 12 passes from naive real-and-imaginary expansion), and polynomial-oscillatory shells ($\deg P = 1$) take 7 rather than 9 real 3-way terms via Proposition 3.4.

Table 2 illustrates concrete operation counts (before an additional 2× reduction in scalar multiplications from even/odd tap-parity folding; 40* denotes the combined pass count for $m = 1$ plus $m = 3$, vs. 48 unshared). On the $N = 161$ ($\omega_c = \pi/64$) orientation-estimation benchmark of [11], replacing direct $2 \times 161^2$ spatial convolution by a two-complex-Gaussian analytical gradient pair reduces arithmetic by 20× (40× with tap-parity folding) without full-image 2-D FFT buffers; in 3-D at $L = 41$, the $r + 3c = 7$ decomposition reduces 68,921 taps/voxel to 861 (80×), saving 22% over naive $1 + 4c = 9$ real/imaginary expansion.

## 6 Formalization Architecture and Verification Audit

Table 3 breaks down the `KernelRankLean` library (5,324 physical lines, 5,021 nonblank lines; earlier stages comprised 3,822 physical lines at commit `6aabb4d` for $d$-D complex and multi-Gaussian ranks, 1,587 lines at commit `cae13d6` for 2-D plus differential annihilators, and 1,340

lines at commit `240dea0` for 2-D classification alone). Three formalization choices proved decisive in Lean 4:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a general dependent family $\prod_{i\in\iota} X_i$ unifies 2-D and $d$-D statements, whereas specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (with a pattern-matched constructor `fin2Vec`) avoids expensive higher-order unification timeouts when rewriting on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.
2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto the finite-dimensional subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting `LinearMap.continuous_of_finiteDimensional` supply automatic continuity.
3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated both coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 3.6 directly.

**Verification audit.** A single script (`scripts/check`, also wired as the repository's pre-commit hook) builds both `KernelRankLean` and `Experiments` via `lake build` (which includes every `.lean` module in the repository), checks the compiler log to ensure no declaration uses `sorry`, runs `#print axioms` on 17 top-level theorems to confirm dependence solely on Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

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This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [14] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [11, 12, 13]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, and suggested extending the formalization to $d$ dimensions. We also thank the aiXiv Official Agent reviewer on version 1.0, whose question regarding higher-degree real polynomial profiles in $d \geq 3$ directly motivated Theorem 3.5 and Proposition 3.4. Both the repository and this manuscript are released under CC0 1.0 Universal.

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